Contextual Learning Automata

Concept Outline and Novelty Assessment

Working Document for PhD Contribution Evaluation

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The Problem: Why Context Matters

Your Current Approach: Global Policy LA

In your current prototype, you have a **single Learning Automaton** that maintains one probability distribution $\mathbf{p} = [p_1, p_2, p_3, p_4]$ over four heuristics (NDVI, FAI, GNDVI, Ensemble).

How it works:

1. For *every* tile, sample an action $a \sim \mathbf{p}$
2. Apply the selected heuristic, compute IoU reward r
3. Update $\mathbf{p}$ based on (a, r)

The limitation: The automaton learns a single "best" heuristic for the *entire dataset*. It cannot adapt to individual tile characteristics.

Why This Matters for Kelp Detection

Different tiles have different characteristics:

Tile Feature	Condition	Likely Best Heuristic
Water turbidity	High (murky)	GNDVI (penetrates better)
Water turbidity	Low (clear)	FAI (designed for surface algae)
Cloud cover	Partial	Ensemble (robust to noise)
Kelp density	Dense canopy	NDVI (strong vegetation signal)
Kelp density	Sparse/submerged	FAI or GNDVI

Key insight: The optimal heuristic depends on the *context* (tile features). A global policy ignores this structure.

Background: Two Separate Fields

Learning Automata (LA)

Learning Automata are a classical approach from the 1970s-80s (Narendra & Thathachar, 1989).

Definition 1 (Learning Automaton)**.** *A Learning Automaton is a tuple* $(\mathcal{A}, \mathbf{p}, T)$ *where:*

- $\mathcal{A} = \{a_1, \ldots, a_n\}$ *is the action set*
- $\mathbf{p} = [p_1, \ldots, p_n]$ *is the probability vector over actions*
- $T : \mathbf{p} \times \mathcal{A} \times \{0, 1\} \to \mathbf{p}'$ *is the update rule*

Key property: Standard LA is *context-free*. The probability distribution $\mathbf{p}$ does not depend on any input features—it's the same for every decision.

Contextual Bandits

Contextual Bandits emerged from the machine learning community (Langford & Zhang, 2007; Li et al., 2010).

Definition 2 (Contextual Bandit)**.** *At each round t:*

1. *Observe context $\mathbf{x}_t \in \mathcal{X}$*
2. *Select action $a_t \in \mathcal{A}$*
3. *Receive reward r_t (only for chosen action)*

The goal is to learn a policy $\pi : \mathcal{X} \to \mathcal{A}$ that maximizes cumulative reward.

Key property: The action selection depends on the context $\mathbf{x}_t$.
Common algorithms: LinUCB, Thompson Sampling, ε-greedy with regression.

The Gap Between These Fields

Property	Learning Automata	Contextual Bandits
Context-aware	No	Yes
Probability-based	Yes (core feature)	Sometimes
Convergence proofs	Strong (stationary env)	Regret bounds
Update mechanism	Simple, local	Often requires optimization
Computational cost	Very low	Can be high
Community	Control theory	Machine learning

Observation: These two fields have developed largely independently. LA researchers don't typically consider context; bandit researchers don't typically use LA-style probability updates.

Has This Been Done Before? Literature Review

Short Answer

Partially, but not in the way I'm proposing.

Related Work

1. Object Migration Automata (OMA)
Oommen and others developed OMA for adaptive list organization. These use context in a limited sense (the object being accessed), but:

- Context is discrete (object identity), not continuous features
- No general feature-based conditioning
- Different problem setting (list organization vs. action selection)

2. Stochastic Point Location Automata
Used for function optimization with continuous action spaces:

- Actions are points in a continuous space

- But no input context—still learning a single optimal point

3. Distributed/Networked Learning Automata
Teams of automata that interact:

- Each automaton is still context-free
- Coordination is via shared rewards, not via context

4. Contextual Bandits with LA-style Updates
To my knowledge: **No established work** directly combines:

- Feature-based context conditioning
- LA probability update rules (L_{R-I}, L_{R-P}, Pursuit, etc.)
- Convergence analysis in the LA tradition

Why Hasn't This Been Done?

1. **Community separation**: LA is from control theory; bandits are from ML. Different conferences, different terminology.
2. **Different goals**: LA focuses on convergence to optimal action; bandits focus on regret minimization.
3. **Computational assumptions**: LA assumes very limited computation (originated for hardware implementation); bandits assume modern computing.

Proposed Formulation: Contextual Learning Automata

Core Idea

Instead of maintaining a single probability vector $\mathbf{p}$, maintain a **function** that maps context to probability:

Definition 3 (Contextual Learning Automaton)**.** *A Contextual Learning Automaton is a tuple* $(\mathcal{A}, \mathcal{X}, \pi_\theta, T)$ *where:*

- $\mathcal{A} = \{a_1, \ldots, a_n\}$ *is the action set*
- $\mathcal{X}$ *is the context space (e.g., tile features)*
- $\pi_\theta : \mathcal{X} \to \Delta^{n-1}$ *is a parameterized policy mapping context to probability simplex*
- T *is a parameter update rule inspired by LA*

Concrete Instantiation: Linear Contextual LA

Let context $\mathbf{x} \in \mathbb{R}^d$ be a feature vector for each tile.

Policy:

$$\pi_\theta(\mathbf{x}) = \text{softmax}(\mathbf{W}\mathbf{x} + \mathbf{b})$$

where $\theta = (\mathbf{W}, \mathbf{b})$ with $\mathbf{W} \in \mathbb{R}^{n \times d}$, $\mathbf{b} \in \mathbb{R}^n$.

Action selection:

$$a_t \sim \pi_\theta(\mathbf{x}_t)$$

Update rule (LA-inspired): Instead of standard gradient descent, use LA-style updates: For L_{R-I} style:

- If reward $r > 0$: update θ to increase $\pi_\theta(\mathbf{x}_t)[a_t]$
- If reward $r \leq 0$: no update (inaction)

Concretely, for the chosen action a_t with context $\mathbf{x}_t$:

$$\mathbf{W}_{a_t} \leftarrow \mathbf{W}_{a_t} + \alpha \cdot r \cdot \mathbf{x}_t \cdot (1 - \pi_\theta(\mathbf{x}_t)[a_t])$$

Alternative: Discretized Context

For simpler analysis, discretize context into clusters:

1. Cluster tiles into K groups based on features (e.g., using k-means)
2. Maintain K separate probability vectors $\mathbf{p}^{(1)}, \ldots, \mathbf{p}^{(K)}$
3. For tile with context $\mathbf{x}$, find cluster $k = \text{cluster}(\mathbf{x})$
4. Use automaton with $\mathbf{p}^{(k)}$

This is simpler to analyze theoretically (it's just K independent automata) but loses the ability to generalize across similar contexts.

What Would Be Novel?

Potential Theoretical Contributions

1. **Convergence analysis for Contextual LA**
   - Under what conditions does the policy π_θ converge?
   - What is the convergence rate?
   - How does it compare to standard contextual bandits?
2. **Regret bounds with LA-style updates**
   - Standard bandits use UCB or Thompson Sampling
   - Can LA-style probability updates achieve sublinear regret?
   - What are the trade-offs?
3. **Analysis of continuous rewards**

- Classical LA assumes binary rewards
- Your setting has continuous rewards (IoU $\in [0, 1]$)
- How do convergence guarantees change?

4. **Non-stationary analysis**
   - Kelp changes seasonally
   - Can we prove adaptation rates under bounded drift?

Potential Practical Contributions

1. **Demonstrate context improves performance**
   - Show that Contextual LA outperforms Global LA on kelp detection
   - Quantify the improvement
2. **Identify which context features matter**
   - Learn which tile features predict optimal heuristic
   - Interpretable model (unlike deep learning)
3. **Efficient online learning**
   - LA updates are computationally simple
   - Important for satellite imagery (large scale)

Honest Assessment: Is This a PhD Contribution?

Arguments FOR

1. **Genuine gap in literature**: LA and contextual bandits have not been systematically combined.
2. **Both theory and practice**: You can contribute both convergence analysis AND practical application to kelp detection.
3. **Novel domain**: Applying LA to remote sensing heuristic selection is itself new.
4. **Interpretability angle**: Unlike deep learning, LA provides interpretable probability dynamics.

Arguments AGAINST / Concerns

1. **"Just combining two things"**: A skeptical reviewer might say you're just putting context into LA, which is "obvious."

   Counter: The theoretical analysis is non-trivial. LA convergence proofs rely on stationary environment and context-free updates. Extending to context requires new proof techniques.
2. **Contextual bandits already exist**: Why not just use LinUCB or Thompson Sampling?

   Counter: LA has different properties (probability-based, simple updates, specific convergence guarantees). The contribution is bringing these properties to the contextual setting.

3. **Limited novelty if only empirical**: If you only show "Contextual LA works better," that's a weak contribution.

   Counter: You need theoretical analysis (convergence, regret bounds) to make it a strong contribution.

My Honest Opinion

This CAN be a PhD contribution, but it requires theoretical depth.

A strong thesis would include:

1. Formal definition of Contextual LA
2. Convergence theorem (under stated assumptions)
3. Regret analysis comparing to standard bandits
4. Empirical validation on kelp detection
5. Demonstration that context actually helps

A weak thesis would be:

1. "I added context to LA"
2. "It works better on my dataset"
3. No theoretical analysis

Alternative Directions (If Contextual LA Doesn't Work)

If after investigation Contextual LA proves too similar to existing work, consider:

Option A: Hierarchical LA

Two-level automaton:

- Level 1: Select detection *scale* (e.g., 10m, 20m, 60m resolution)
- Level 2: Select heuristic at that scale

Novel: Theoretical analysis of hierarchical LA convergence.

Option B: Non-Stationary LA with Drift Detection

Focus on the temporal aspect:

- Kelp appearance changes seasonally
- Design LA that detects distribution shift and adapts

Novel: Regret bounds under bounded non-stationarity.

Option C: Spatially-Aware LA

Focus on the spatial aspect:

- Neighboring tiles share information
- Graph-regularized LA

Novel: Convergence analysis with spatial correlation.

Concrete Next Steps

If you want to pursue Contextual LA:

1. **Literature deep-dive**: Search specifically for "contextual learning automata," "feature-dependent LA," "state-dependent LA." Confirm the gap.
2. **Implement discretized version**: Cluster tiles, run separate automata per cluster. See if context helps empirically.
3. **Formalize the theory**: Write down the convergence theorem you want to prove. Identify the assumptions needed.
4. **Discuss with advisor**: Get feedback on whether this direction is publishable and sufficient for PhD.

Open Questions for You to Consider

Open Question 1. *What tile features would you use as context? (Water indices, cloud mask values, geographic location, season?)*

Open Question 2. *How many distinct "contexts" exist in your data? Is discretization feasible?*

Open Question 3. *Are you comfortable with theoretical proof work, or do you prefer empirical focus?*

Open Question 4. *What does your advisor think about the LA direction generally?*